How-To Library

Informatica

Enable Customer Managed Keys for MDM SaaS on Microsoft Azure

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Abstract

This article explains how to create your own native master encryption key for MDM SaaS on Microsoft Azure. You can control and maintain the key that you create. Use the key to control and restrict access to your organization's data. MDM SaaS leverages the keys that you control to encrypt data at rest, including data that persist in the data store, search indexes, and index backup.

Supported Versions

- MDM SaaS November 2023 or later

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Overview

IDMC protects your organization's sensitive data in the cloud using organization-specific encryption keys that are generated and stored in the IDMC key management service (KMS). To prevent malicious access, the keys are encrypted using a master key that is stored in the cloud provider's KMS.

If you prefer, you can create a customer-managed key (CMK). When you create a CMK, you control access to it. However, you'll need to grant IDMC access to the CMK so that it can encrypt and decrypt your organization's sensitive data.

Creating a CMK offers the following benefits:

- You can restrict and control any access to your data.
- You can restrict the decryption of your data in emergency situations.
- You create and hold the key material in your KMS. The key is never exposed to your cloud service provider.
- You maintain full control of the key throughout its lifecycle. You can revoke access or delete the key at any time.

Note: When you create a CMK, your KMS and IDMC POD must use the same cloud provider. For example, if your IDMC POD is on a Microsoft Azure POD, then you must store your CMK in Azure Key Vault. You can't store it in AWS KMS or Google Cloud KMS.

Note: Your KMS must be deployed in the same region as the Informatica Intelligent Cloud Services POD, for both primary and disaster recovery environments. To identify the primary and disaster recovery regions for your organizations, see the POD Availability and Networking page.

The instructions related to Microsoft Azure in this article are for reference purposes. For more detailed information about Microsoft Azure, see the Microsoft Azure documentation.

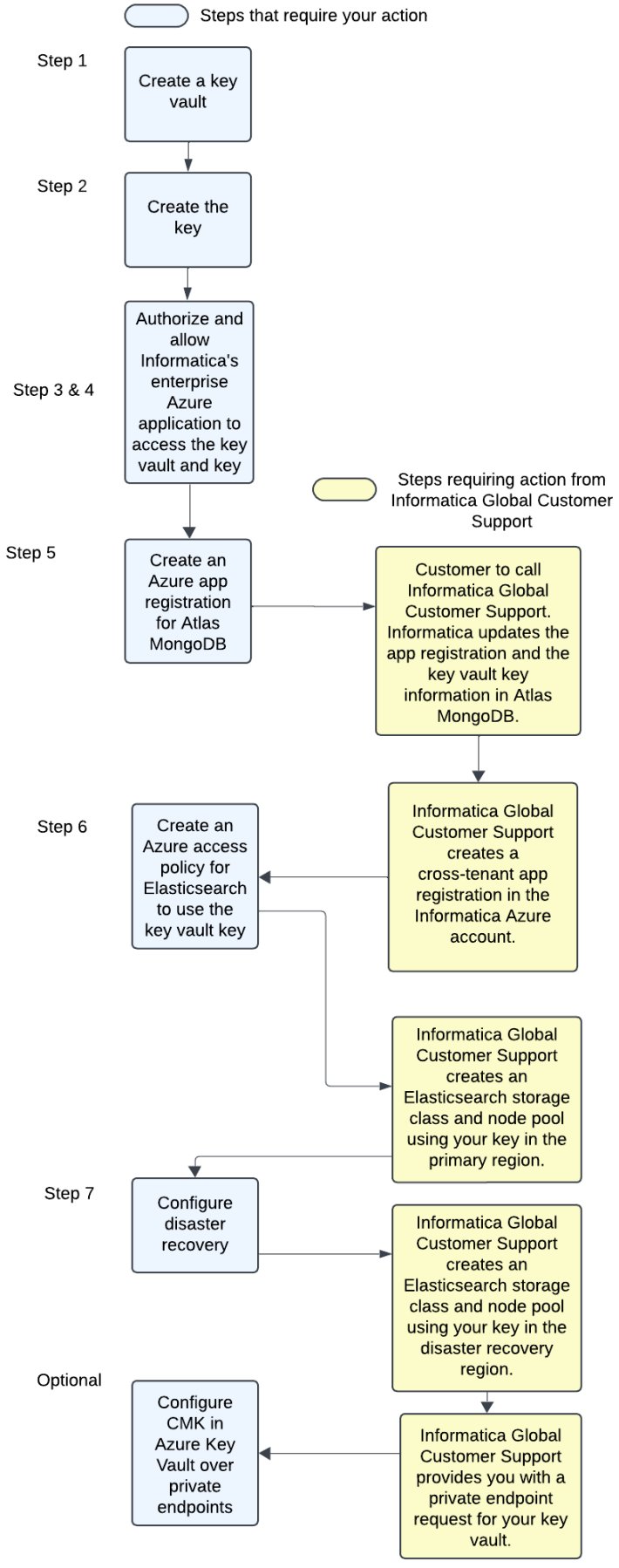
Steps to create and enable the key

To create and use a CMK, you provision the key in Azure Key Vault and enable cross-account access with IDMC. Then you enable customer managed keys in IDMC. The configuration is based on an MDM SaaS single tenant with your CMK.

To create and enable a CMK, complete the following steps:

1. In Azure Key Vault, create a key vault.
2. In the key vault that you created, generate a key to use as your CMK.
3. Authorize Informatica's enterprise Azure application to access the key.

   Note: You must repeat this step for all your organizations.
4. Create an app registration for the key to use in MongoDB and provide your Azure subscription account and key vault credentials to Informatica Global Customer Support.
5. Obtain cross-tenant application details from Informatica Global Customer Support to create an access policy for Elasticsearch and manage key vault secrets and keys.
6. Configure a disaster recovery region key vault.
7. Optionally, to encrypt your data, configure CMK in Azure Key Vault over private endpoints.


Steps that require your action
Step 1
Create a key vault
Step 2
Create the key
Step 3 & 4
Authorize and allow Informatica's enterprise Azure application to access the key vault and key
Steps requiring action from Informatica Global Customer Support
Step 5
Create an Azure app registration for Atlas MongoDB
Customer to call Informatica Global Customer Support. Informatica updates the app registration and the key vault key information in Atlas MongoDB.
Informatica Global Customer Support creates a cross-tenant app registration in the Informatica Azure account.
Step 6
Create an Azure access policy for Elasticsearch to use the key vault key
Informatica Global Customer Support creates an Elasticsearch storage class and node pool using your key in the primary region.
Step 7
Configure disaster recovery
Informatica Global Customer Support creates an Elasticsearch storage class and node pool using your key in the disaster recovery region.
Optional
Configure CMK in Azure Key Vault over private endpoints
Informatica Global Customer Support provides you with a private endpoint request for your key vault.

Step 1. Create a key vault

In Azure Key Vault, create a key vault to store your CMK. Note the key vault URI because you need it when you allow Informatica's application to use the key vault. Ensure that the key vault you create is in the same Azure region that IDMC uses.

1. Log in to the Azure portal.
2. In the **Search** box, enter `Key Vault`.
3. From the results list, select **Key Vault**.
4. In the **Key Vault** section, select **Create**.
5. To configure the key vault details, perform the following taks:
   a. Select the same Azure region that IDMC use for the key vault.
   b. Enable soft delete and purge protection.

      Note: If the key vault is for a disaster recovery region, enable soft delete and disable purge protection so that you can delete and purge the key and restore it from the key in the primary region.

   c. Click **Next**.
6. On the **Access configuration** tab, select the permission model, review and update the other access policies as needed, and click **Next**.

   You can choose either the **Azure role-based access control (recommended)** or **Vault access policy** permission model.

   Note: Azure role-based access control (Azure RBAC) is the default and recommended access control model for Azure Key Vault. The Vault access policy model is deprecated and is planned to be retired in February 2027. For more information, see Migrate to Azure RBAC from access policies in the Key Vault documentation.
7. On the **Networking** tab, review the networking details and click **Next**.
8. On the **Tags** tab, add tags to the key vault and click **Next**.
9. On the **Review + create** tab, review the key vault configuration and click **Create**.
10. To view the created key vault, click **Go to Resource**, and note the vault URI.

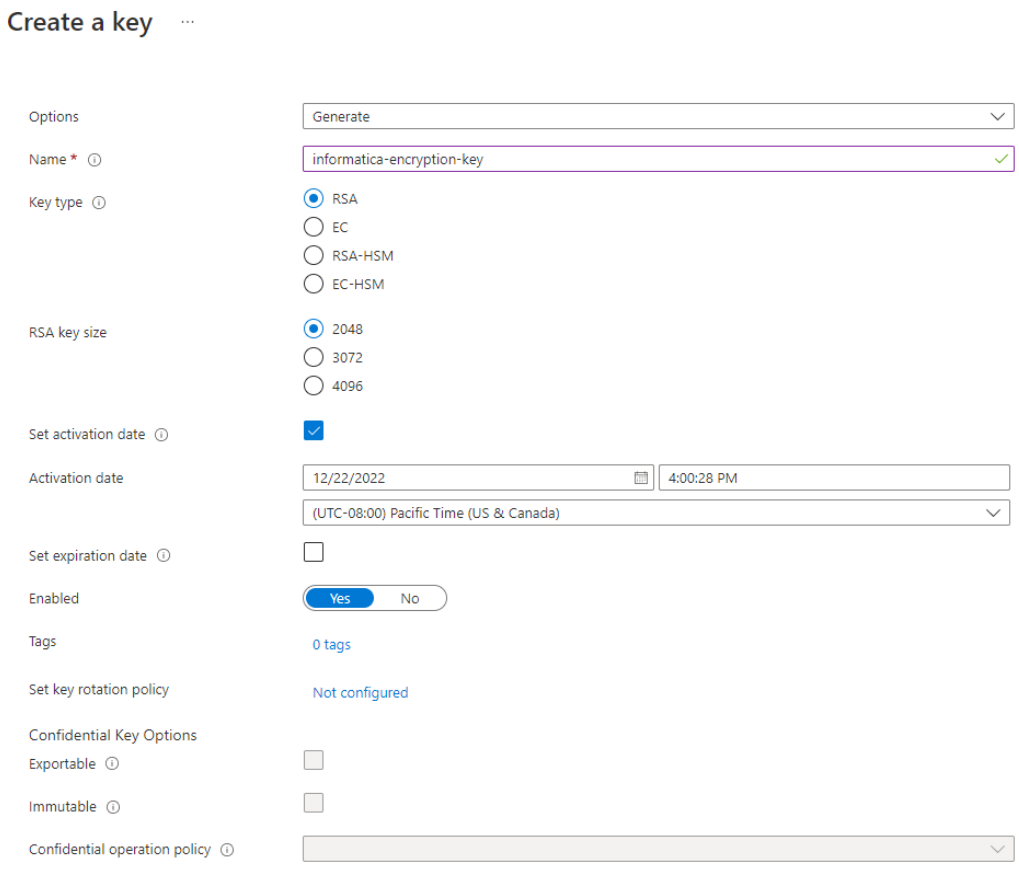
Step 2. Create the key

In the key vault that you created, create a symmetric key to use as your CMK. Note the key name and version because you need them when you allow Informatica's application to use the key vault.

1. In the **Search** box, enter **Key Vault**.
2. From the results list, select the key vault that you created in "Step 1. Create a key vault" on page 5.
3. In the left panel under **Objects**, select **Keys**.
4. Select **Generate/Import**.
5. Select **Generate** from the drop-down list, and configure the following options:

Property	Value
Name	Key name, for example, `informatica-encryption-key`
Key type	RSA

Property	Value
RSA key size	Choose any key size.
Set activation date	Enabled
Activation date	Today's date and time
Enabled	Yes
Set key rotation policy	Optionally, configure a key rotation policy.

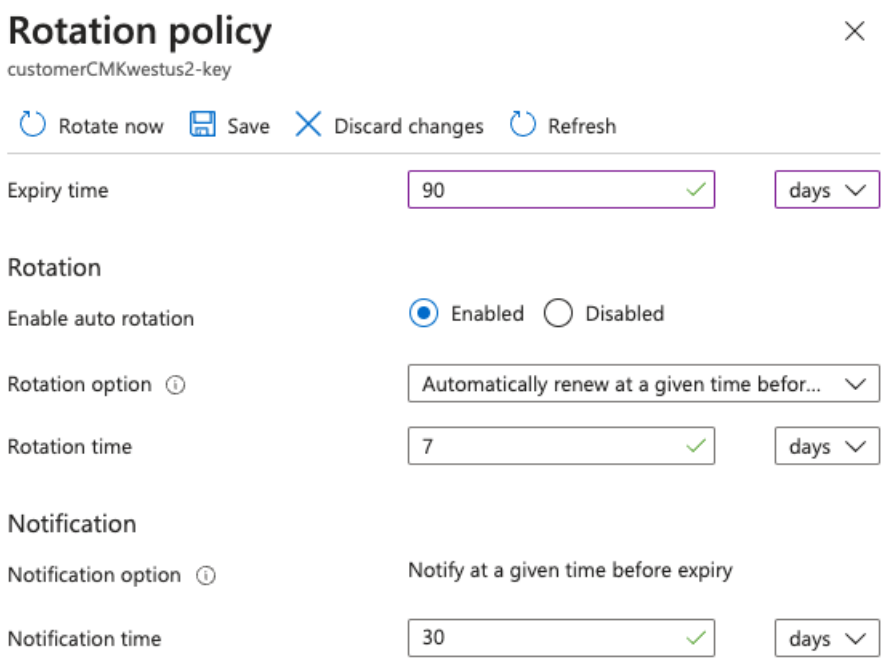


6. To add rotation policy details, enter the following values:

   Note: Ensure that the expiry time is set to a minimum of 90 days.

Property	Value
Expiry time	90 days
Enable auto rotation	Enabled

Property	Value
Rotation option	Automatically renew at a given time before expiry
Rotation time	7
Notification time	30 days



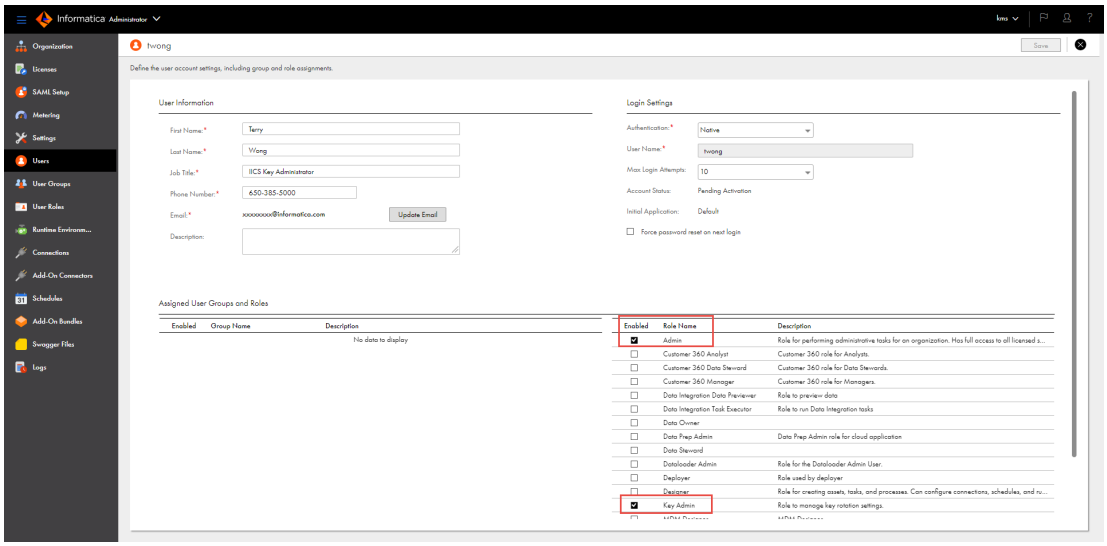
7. Click **Create**.
8. Click the key and note the key name and version.

Step 3. Authorize Informatica's enterprise Azure application to access the key

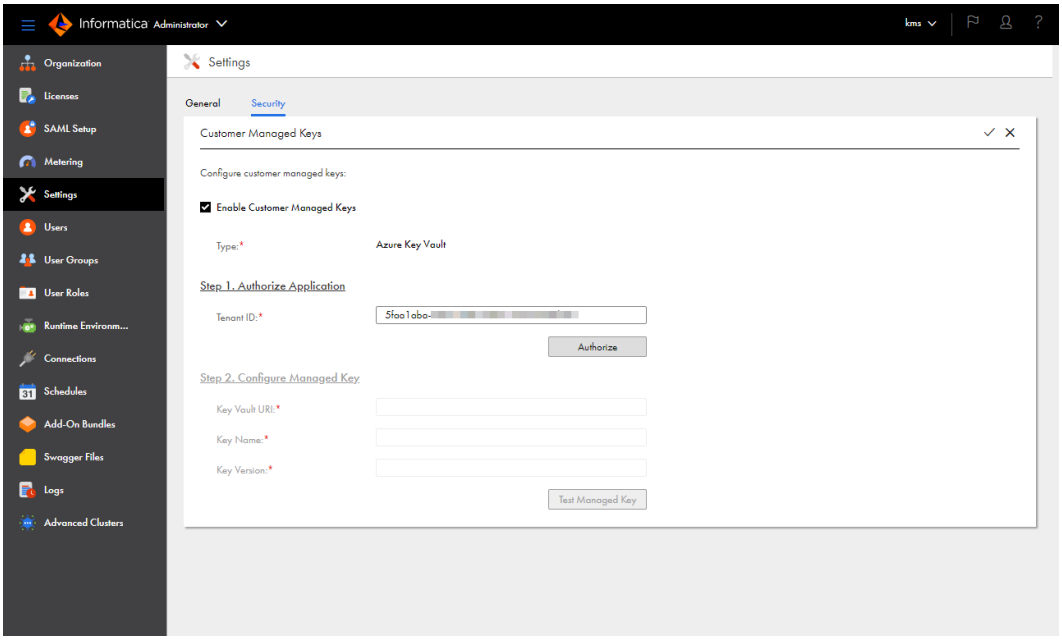
In IDMC Administrator, open the **Settings** page and enable customer managed keys for your organization. Then, grant the Informatica KMS Connect app permission to access your CMK.

Note: You must repeat this step for all your organizations.

Note: Before you can complete this step, you need to assign at least one administrative user the **Admin** and **Key Admin** roles on the user details page in Administrator:



1. Log in to IDMC Administrator with a user account that has both the Admin and Key Admin roles.
2. Open the **Settings** page and click the **Security** tab.
3. Click the edit (pencil) icon.
4. Select **Enable Customer Managed Keys**.
5. In **Step 1. Authorize Application**, enter the **Tenant ID** of your Azure Subscription.



Tip: You can find the Tenant ID under **Properties** in Azure Active Directory.

6. Click **Authorize**.
7. When prompted, sign in to your Azure account.

   Note: You must sign in with a user account that has the Cloud Application Administrator role.

8. In the **Permissions requested** dialog box, click **Accept** to grant permissions to the Informatica KMS Connect app.

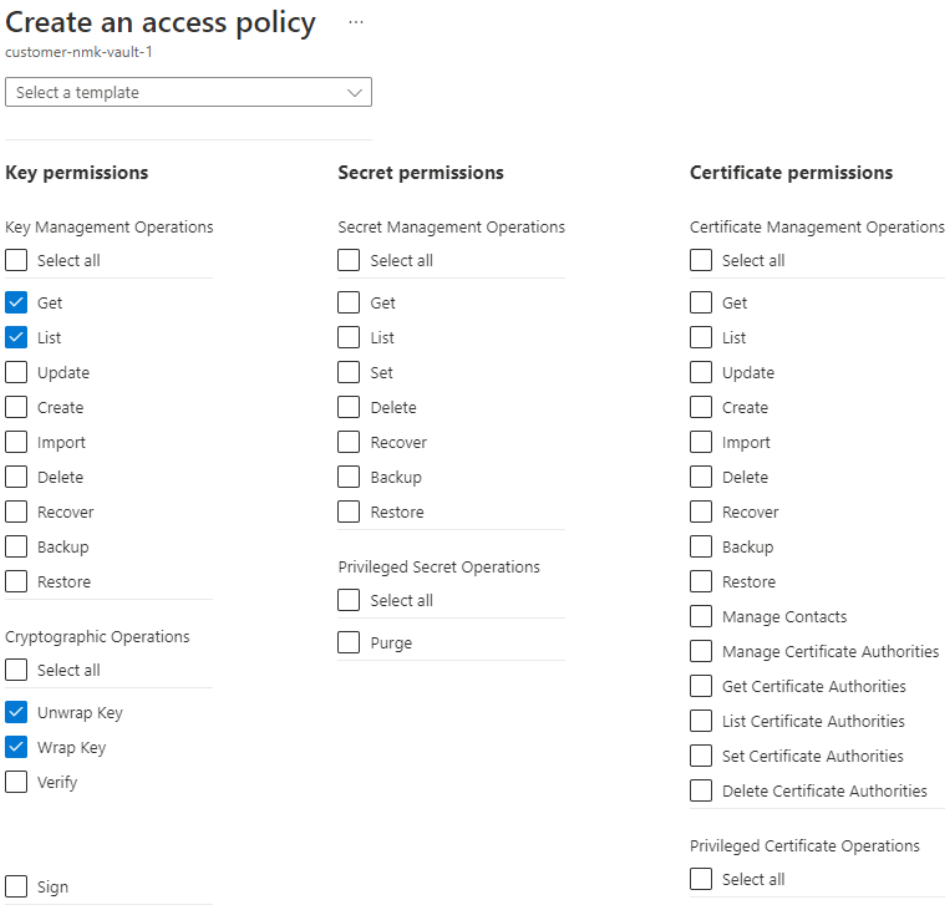
Step 4. Allow Informatica's enterprise Azure application to use the key vault

Allow the Informatica KMS Connect app to use the Azure AD application credentials and tenant ID. To do this, first create an access policy for the Informatica KMS Connect app in Azure Key Vault, and then enter the key information on the **Settings** page in IDMC.

Note: You must repeat this step for all your organizations.

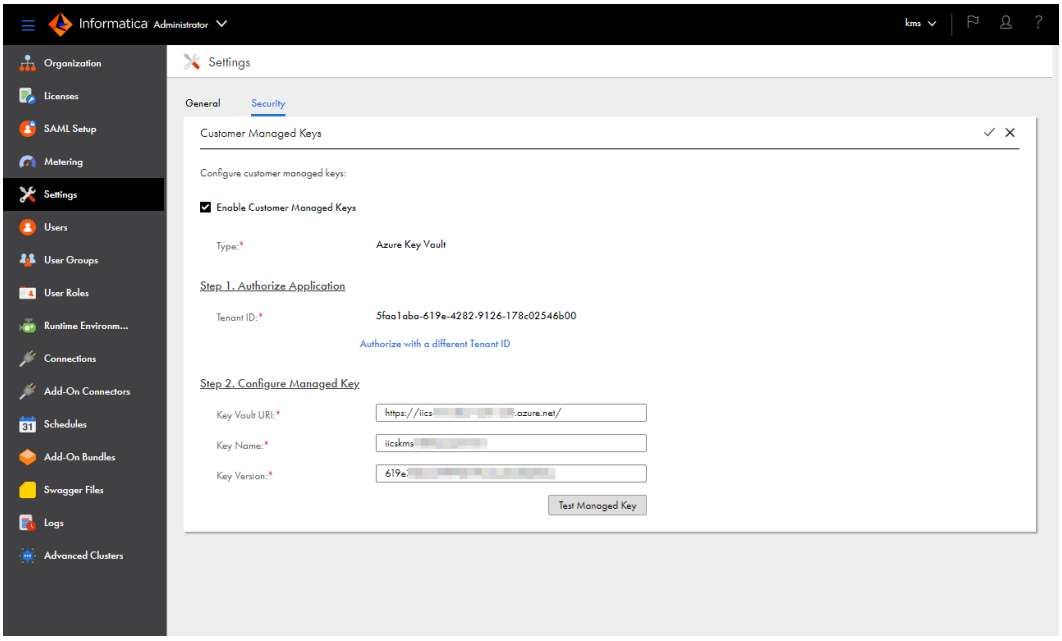
1. Log in to the Azure portal.
2. In the **Search** box, enter **Key Vault**.
3. From the results list, select **Key Vault**.
4. Select the key vault you created in "Step 1. Create a key vault" on page 5.
5. Grant the following permissions:
   - Get
   - List
   - Decrypt
   - Encrypt
   - Unwrap Key
   - Wrap Key

The following image shows the permissions to select:



6. Log in to IDMC Administrator with a user account that has both the Admin and Key Admin roles.
7. Open the **Settings** page and click the **Security** tab.
8. Click the edit (pencil) icon.

9. In **Step 2. Configure Managed Key**, enter the key vault URI, key name, and key version.



10. To test the key, click **Test Managed Key**.

    A success message appears if the test was successful.

11. Click the save (check mark) icon to save your changes.

Step 5. Enable encryption at rest using your Azure key management in MongoDB Atlas

To enable encryption at rest using Azure key management in MongoDB Atlas, configure an Azure service principal, a key vault access policy, and role-based access. Then, provide Azure subscription ID, service principal, and key vault details to Informatica Global Customer Support. Informatica uses these details to configure encryption at rest in MongoDB Atlas.

1. Run the following Azure CLI command to log in to your Azure subscription:

   ```
   az login
   ```

2. To retrieve the tenant ID, run the following command:

   ```
   az account show --query tenantId
   ```

3. To create a service principal, verify that your account has the required Microsoft Entra ID role.
   a. Log in to the Azure portal, navigate to **Microsoft Entra ID**, and click **Roles and administrators**.
   b. Verify that your account has the Application Developer role or one of the following higher-privilege roles:
      - Application Administrator
      - Cloud Application Administrator
      - Global Administrator
      - Hybrid Identity Administrator

      If your account doesn't have one of these roles, contact your Azure administrator to have one assigned before you continue.

4. Create a service principal for MongoDB Atlas.
   a. To check if a service principal for MongoDB Atlas already exists in your subscription, run the following command:

      ```
      az ad sp list --filter "AppId eq '9efedfcc-2eca-4b27-a613-0cad1e114cb7'"
      ```

      The value `9efedfcc-2eca-4b27-a613-0cad1e114cb7` is the MongoDB Atlas application ID.
   b. If the command returns an empty result, run the following command to create a new service principal:

      ```
      az ad sp create --id 9efedfcc-2eca-4b27-a613-0cad1e114cb7
      ```
   c. Note the object ID that you can use to grant the service principal access to your key vault.
5. To grant the service principal access to your key vault, perform the following steps:
   a. To assign the Reader role to allow MongoDB Atlas to read the key vault resource, run the following command:

      ```
      az role assignment create \  --assignee-object-id "<SERVICE_PRINCIPAL_ID>" \  --
      assignee-principal-type ServicePrincipal \  --role "Reader" \  --scope "/
      subscriptions/<SUBSCRIPTION_ID>/resourceGroups/<RESOURCE_GROUP>/providers/
      Microsoft.KeyVault/vaults/<KEY_VAULT_NAME>"
      ```

      The SERVICE_PRINCIPAL_ID is the object ID you retrieved or created in Step 4.con page 12.
   b. To grant the service principal key permissions using a key vault access policy, run the following command:

      ```
      az keyvault set-policy \  --name "<KEY_VAULT_NAME>" \  --object-id
      "<SERVICE_PRINCIPAL_ID>" \  --key-permissions get encrypt decrypt
      ```
6. Contact Informatica Global Customer Support in the following scenarios:
   - You configured your CMK in Azure Key Vault over a private endpoint and need to remove the private endpoint connection.
   - Your service principal configuration is complete and you are ready for Informatica to update the encryption at rest settings in MongoDB Atlas.

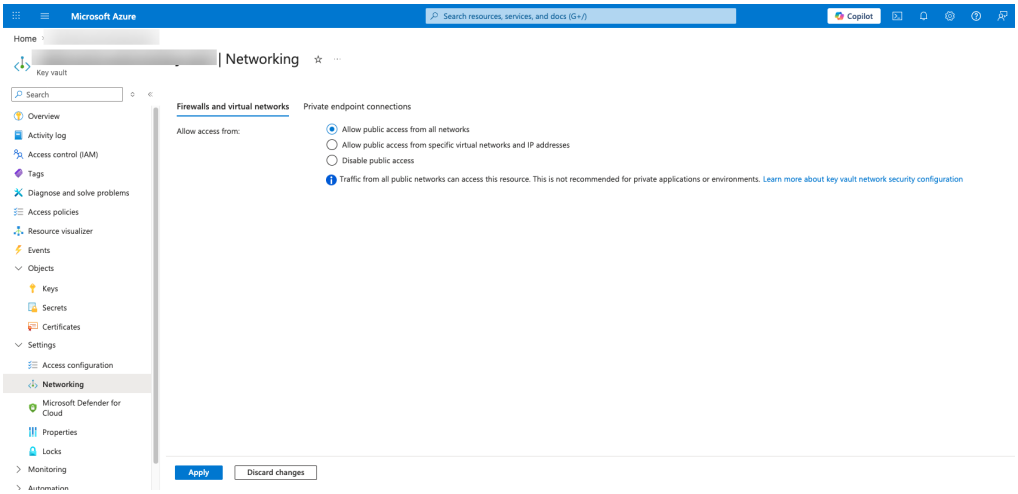
   Provide Informatica Global Customer Support the following details:
   - Azure tenant ID
   - Service principal object ID
   - Confirmation that the `az role assignment` and `az keyvault set-policy` commands ran successfully
   - Azure subscription ID for the key vault
   - Resource group name for the key vault
   - Key vault name
   - Key identifier

   Informatica Global Customer Support updates the encryption at rest settings in MongoDB Atlas using the service principal and notifies you when the update is complete.
7. If your key vault restricts public access by network or IP address, perform the following steps to temporarily allow access from all networks for MongoDB Atlas validation:
   a. Log in to the Azure portal and navigate to your key vault.
   b. Click **Networking** and click the **Firewalls and virtual networks** tab.
   c. Note the following key vault network settings before you make any changes:
      - Firewall rules
      - Virtual network rules

- Exceptions

d. Select **Allow public access from all networks** and click **Apply**.

The following image shows the settings to select:



8. To restore your private endpoint connection to your Azure Key Vault, contact Informatica Global Customer Support.
9. To restore your Azure key vault network settings to the original configuration, perform the following steps:
   a. Under **Firewalls and virtual networks**, select **Allow public access from specific virtual networks and IP addresses**.
   b. Restore the virtual network and IP address rules that you noted in Step 3, and click **Apply**.

   For more information about managing Azure service principal access, see https://www.mongodb.com/docs/atlas/security/set-up-azure-access/?interface=atlas-ui.

Step 6. Create Azure access policy for Elasticsearch to use key vault key

Obtain cross-tenant application details from Informatica to create an access policy for Elasticsearch.

1. Log in to the Azure portal.
2. Click the **Cloud Shell** icon located next to the search bar.
3. Run the following Azure CLI command to add the service principle provided by Informatica:

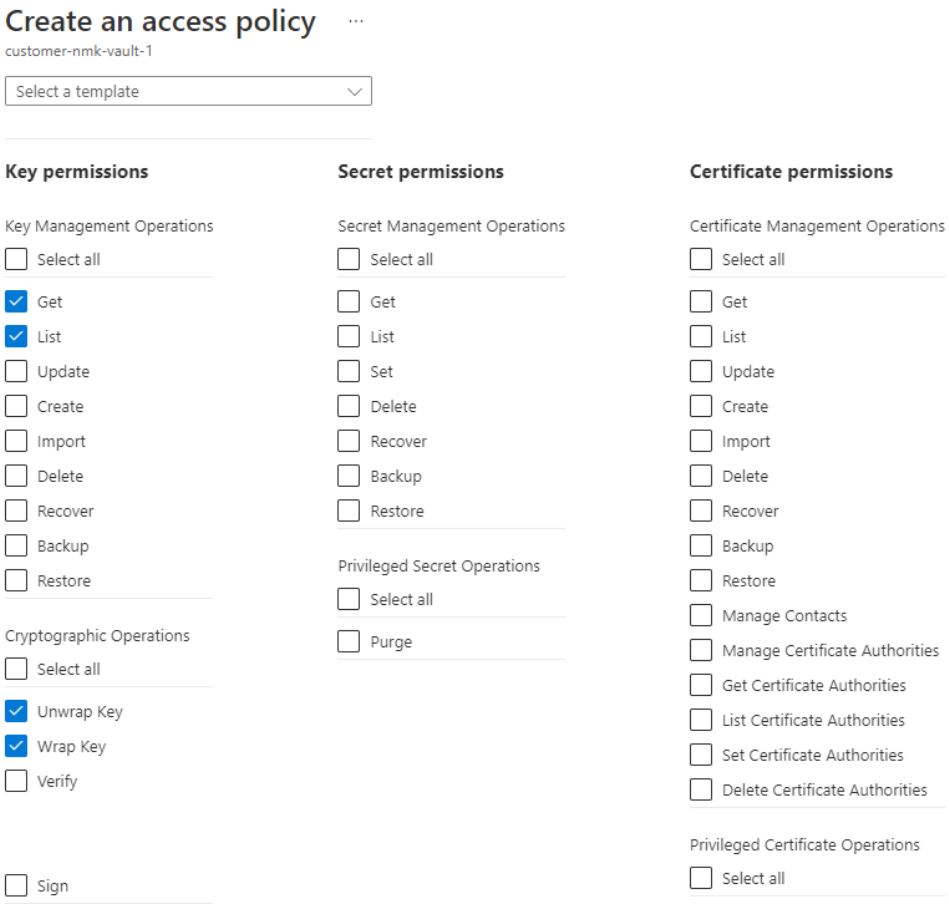
```
#Create the service principal with the registered app's application ID (client ID)
multiTenantAppId="<multi-tenant-app-id provided by Informatica>"

az ad sp create --id $multiTenantAppId --query id --out tsv
```

4. In the **Search** box, enter **Key Vault**.
5. From the results list, select **Key Vault**.
6. Select the key vault you created in "Step 1. Create a key vault" on page 5.
7. Grant the following permissions:
   - Get
   - List

- Decrypt
- Encrypt
- Unwrap Key
- Wrap Key

The following image shows the permissions to select:



8. Under **Principal**, search for "`<multi-tenant app provided by Informatica>`" and select it.
9. After you create the access policy, notify Informatica to create a storage class and node pool in the primary region.

Step 7. Configure disaster recovery

Informatica offers a dedicated recovery region for use in case of a disaster. To configure the disaster recovery region, create a new key vault and key. Informatica provides you with the cross-tenant application necessary to set up the key vault in the recovery region. You need to create an Azure access policy to allow Elasticsearch to access the key vault key.

1. Create a new key vault in the disaster recovery region defined by Informatica.

   For more information about creating a key vault, see "Step 1. Create a key vault" on page 5.
2. Create a new key in the disaster recovery region.

   For more information about creating a key, see "Step 2. Create the key" on page 5.

3. Assign the roles you've assigned in the primary region to access Azure app registration for Atlas MongoDB for the new key in the disaster recovery region.
    1. Navigate to **Resource Group**.
    2. Click **Access control (IAM)**.
    3. Click **Add a role assignment**.
    4. Select **Role** as **Reader**.
    5. Select **Assign access to** as **Azure AD user, group** or **service principal**.
    6. Select the name of the app, and click **Save**.

    Informatica updates MongoDB using the URL for the disaster recovery region key during a disaster recovery. MongoDB continues to use the same app registration secret as the primary region key for the disaster recovery region key.
4. Create an Azure access policy for the new key.

    Informatica provides the cross-tenant application for key vault access in the disaster recovery region.

    For more information about creating Azure access policy, see "Step 6. Create Azure access policy for Elasticsearch to use key vault key" on page 13.
    Provide the disaster recovery region key information to Informatica. Informatica requires the disaster recovery region key to configure Elasticsearch before a disaster.

When IDMC is available during disaster recovery, authorize and allow the IDMC to access the disaster recovery region key vault key.

For more information about authorizing and allowing Informatica's enterprise Azure application, see "Step 3. Authorize Informatica's enterprise Azure application to access the key" on page 7 and "Step 4. Allow Informatica's enterprise Azure application to use the key vault" on page 9.

Optionally configure a CMK in Azure Key Vault over private endpoints

To encrypt your data at rest in Atlas MongoDB, you can configure your CMK from Azure Key Vault over private endpoints. Additionally, you can configure traffic to your Azure Key Vault through Azure Private Link.

1. Ensure that you use your Azure Key Vault and the app registration secret to encrypt the MongoDB cluster.
2. Select one of the following public access options for the Azure Key Vault network:
    - Allow public access from specific virtual networks and IP addresses
    - Allow public access from all networks
3. Ensure that you register your Azure subscription for the resource provider in the Microsoft Network.

    For more information, see https://learn.microsoft.com/en-us/azure/azure-resource-manager/management/resource-providers-and-types.
4. Contact Informatica Global Customer Support and provide Azure Key Vault details required for configuring Atlas MongoDB.
5. After Informatica completes the Atlas MongoDB configuration, contact Informatica Global Customer Support to obtain private endpoints for the Azure Key Vault in the primary and disaster recovery regions.

    For more information about adding Informatica IP addresses to the vault's allowlist, see *Enable Customer Managed Keys for your Organization on Microsoft Azure*. The IP addresses for your POD are listed on the POD Availability and Networking page.
6. Verify and approve the private endpoint connection for the primary and disaster recovery regions.

Informatica Global Customer Support validates the connection.

7. To enhance the security of your Azure Key Vault, select one of the following options:
   - Disable public access or Allow public access from specific virtual networks and IP addresses.
   - Allow trusted Microsoft services to bypass this firewall.

Frequently asked questions

When I clicked **Test Managed Key** in IDMC, the test failed. What should I do?

If you get an error when testing the key, perform the following checks:

- In IDMC Administrator, verify that the key settings on the **Settings** page match the settings for the CMK in the Azure portal.
- In the Azure portal, verify that the status of the CMK is active.
- In the Azure portal, verify that the permissions on the CMK allow Informatica cryptographic access to the key.

If you continue to encounter errors, contact Informatica Global Customer Support.

What happens if the CMK is rotated in Azure Key Vault?

You can rotate the CMK in Azure Key Vault manually or on a schedule. Rotating a key creates a new version. The old version remains in Azure Key Vault and is used for decryption only.

IDMC, Elasticsearch, and MongoDB Atlas automatically detect key rotation. When the CMK is rotated in Azure Key Vault, the old CMK decrypts your encryption keys, and the new CMK re-encrypts them.

What if I need to update the CMK in Azure Key Vault?

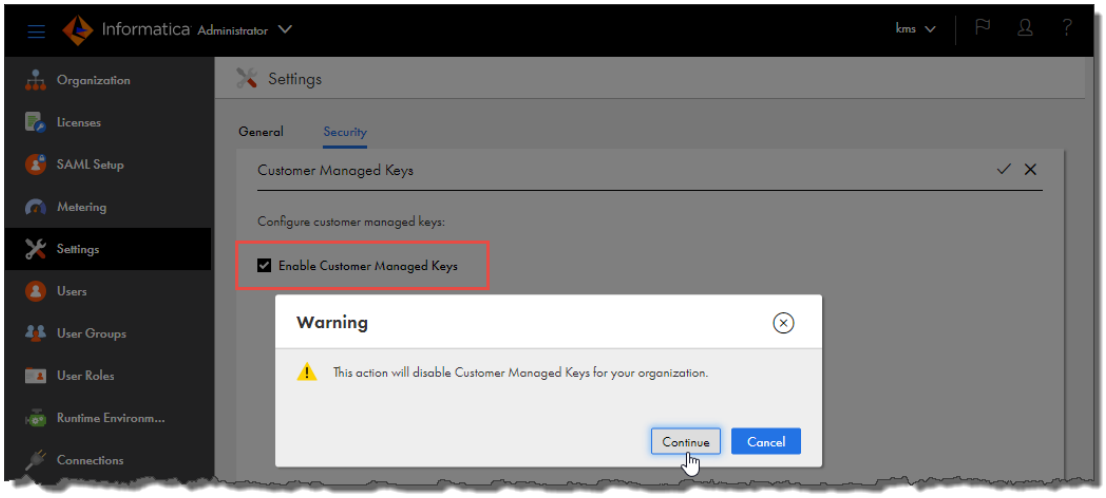
If you need to update the CMK, first provision a new CMK in Azure Key Vault. Then, update the key details on the **Settings** page in IDMC Administrator.

Note: Be sure to keep the old version of the CMK in Azure Key Vault active until you update the key details in IDMC.

You can delete the old version of the CMK in Azure Key Vault after you update the key details on the **Settings** page in IDMC Administrator.

What if I want Informatica to manage key encryption?

If you want Informatica to manage key encryption, you can disable the **Enable Customer Managed Keys** option on the **Settings** page in IDMC Administrator:



When you do this, be sure to keep the current version of the CMK in Azure Key Vault active. If the CMK is not active, disabling customer managed keys in IDMC fails.

When you disable this option, your organization's encryption keys are once again encrypted using encryption keys that are managed by Informatica. It can take up to 10 minutes for the Informatica encryption keys to become active.

You can disable or delete the CMK in Azure Key Vault after you disable the **Enable Customer Managed Keys** option in Administrator.

What if I want to temporarily revoke Informatica's access to the CMK?

If you want to temporarily revoke Informatica's access to the CMK, you can disable the key in Azure Key Vault.

When you disable the CMK, IDMC can no longer unencrypt your organization's encrypted data, and any jobs that use the data will fail until you reactivate the CMK in Azure Key Vault.

How do I replace the CMK if I suspect it has been compromised?

If you want to replace the CMK, you can delete the key in Azure Key Vault and create a new one.

Warning: Deleting the CMK in Azure Key Vault results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you need to replace the CMK, perform the following steps so that you don't lose access to the encrypted data and jobs don't fail:

1. In Administrator, open the **Settings** page, click the **Security** tab, and disable the **Enable Customer Managed Keys** option.
2. In the Azure portal, delete the CMK.
3. In the Azure portal, create a new CMK.
4. On the **Settings** page in IDMC Administrator, re-enable the **Enable Customer Managed Keys** option and enter the details for the new CMK.

Can I delete the CMK if I don't want Informatica to access any of my encrypted data?

Warning: Deleting the CMK in Azure Key Vault results in permanent loss to any encrypted data in IDMC and causes the jobs that use the data to fail.

If you're sure that you want Informatica to forgo all access to your encrypted data in IDMC, you can delete the CMK in Azure Key Vault.

Can I use separate encryption keys for Development, Production, and QA organizations?

Informatica requires you to use a single master key for the deployed organizations. MDM SaaS leverages the master key to encrypt unique tenant-specific keys, ensuring that data isolation and security are maintained for each tenant individually.

Can I request my organizations to host on different PODs?

When you use Customer Managed Keys, all organizations must host within a single POD.

Author

Informatica Documentation Team